

Math 534H - Spring 2026 - Lecture 20

①

Recall from last time that for any $n \geq 1$ the function

$$u_n(x, t) = \left[\tilde{A}_n \cos\left(\frac{n\pi}{L} ct\right) + \tilde{B}_n \sin\left(\frac{n\pi}{L} ct\right) \right] \sin\left(\frac{n\pi}{L} x\right)$$

where $\tilde{A}_n, \tilde{B}_n$ are arbitrary constants is a solution to the problem:

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0, & 0 < x < L, \quad t > 0 \\ u(0, t) = 0 = u(L, t), & \forall t \geq 0 \end{cases}$$

where we only took into account the wave equation and the zero Dirichlet boundary conditions

but not yet the IC:

(2)

$$u(x, 0) = \phi(x) \quad \text{and} \quad u_t(x, 0) = \psi(x).$$

We found $u_n(x, t)$ by separation of variables solving two eigenvalue problems (see page 2, Lecture 19)

finding $\lambda = \lambda_n = \beta_n^2 := \left(\frac{n\pi}{L}\right)^2$ which are called the eigenvalues. In solving these two eigenvalue

problems we made the a-priori assumption that

$\lambda > 0$. We still need to prove this is the case
(We'll do a bit later).

We then turn to the question of when and how

such $u_n(x, t)$ or any superposition of these

$$u_N(x, t) = \sum_{n=1}^N u_n(x, t) \text{ satisfy also}$$

the given initial conditions. That is we need to know when

$$\textcircled{1} u_N(x, 0) = \sum_{n=1}^N \tilde{A}_n \sin\left(\frac{n\pi}{L}x\right) \stackrel{?}{=} \phi(x) \quad \downarrow \text{given!}$$

$$\textcircled{2} \left(u_N\right)_t(x, 0) = \sum_{n=1}^N \left(\tilde{B}_n \frac{n\pi}{L} \cdot c\right) \sin\left(\frac{n\pi}{L}x\right) \stackrel{?}{=} \psi(x) \quad \uparrow$$

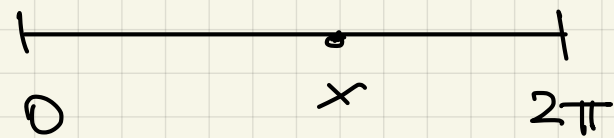
Q: Can we find $\tilde{A}_n, \tilde{B}_n$ constants such that this is true? For which type of functions

ϕ, ψ is this true?

④

We saw that for example if $\phi(x) = \sin x$, $L = 2\pi$

then $\tilde{A}_n = \begin{cases} 1 & \text{if } n=2 \\ 0 & \text{if } n \neq 2 \end{cases}$



satisfy ①

And if $\psi(x) = \sin(4x)$, $0 < x < 2\pi$, then

if we call $\tilde{D}_n = \tilde{B}_n \frac{\pi n}{L} c$ we have that

$$\tilde{D}_n = \begin{cases} 1 & \text{if } n=8 \\ 0 & \text{if } n \neq 8 \end{cases} \quad \text{satisfy } ②$$

But these ϕ, ψ are very special functions!

⑤

Remark: Recall $\lambda_n = \left(\frac{\pi n}{L}\right)^2$ are called the eigenvalues

The numbers $\left(\frac{\pi n}{L}\right)$ themselves are called the frequencies of the solution.

To understand how to answer in general the question about when and how to also satisfy the initial conditions we'll need to

study some Fourier analysis and Fourier series for periodic functions (Chapter 5).

We will see that one can prove that ANY

CONTINUOUS functions ϕ and ψ can be represented ^{L -periodic} as a (possibly infinite) linear combination of sine sums (SINE SERIES) or of cosine sums (COSINE SERIES). (6)

(by infinite/series we mean N must be allowed to go to ∞).

We turn now to the Heat/Diffusion Eq. on $[0, L]$ with zero Dirichlet BC.

The BIVP ($k > 0$) (7)

$$\begin{array}{l} \text{EQ} \\ \text{BC} \\ \text{IC} \end{array} \left\{ \begin{array}{l} u_t - k u_{xx} = 0 \quad 0 < x < L, t > 0 \\ u(0, t) = 0 = u(L, t) \quad \forall t \\ u(x, 0) = \phi(x) \quad x \in [0, L] \end{array} \right.$$

$\hookrightarrow$ given

We proceed as before. We look for solutions of the form $u(x, t) = X(x)T(t)$

We plug into the EQ to get $\begin{pmatrix} X(x) \neq 0 \\ T(t) \neq 0 \end{pmatrix}$

$$\frac{T'(t)}{k T(t)} = \frac{X''(x)}{X(x)} = \text{CONSTANT}$$

only dep. on t only depends on x (in x and t)

Same as before, let's call this constant $-\lambda$ ⑧

and claim (we have to prove still) $\lambda > 0$

Assuming the claim $\lambda = \beta^2$ we set

$$\text{2nd order ODE} \left\{ \begin{array}{l} X''(x) + \beta^2 X(x) = 0 \end{array} \right.$$

$$\text{1st order ODE} \left\{ \begin{array}{l} \underline{T'(t) + \beta^2 k T(t) = 0} \end{array} \right.$$

Note that again the zero Dirichlet BC

affect X

$$X(0)T(t) = 0 = X(L)T(t) \quad \forall t$$

$$\Rightarrow X(0) = 0 = X(L) \quad \underline{\text{same as before}}$$

⑨

For $X(x)$ then we have the same ODE with the same BC. we know what $X(x)$ is. and also that as before $\beta = \frac{n\pi}{L}$, $n \geq 1$
(so $\lambda = \beta^2 = \left(\frac{n\pi}{L}\right)^2$ eigenvalues)

So let's focus on $T(t)$. we set $\left(\beta = \frac{n\pi}{L}\right)$

$$T(t) := A_n e^{-\beta^2 k t} \quad \left(\text{decaying exponential as } t \rightarrow \infty\right).$$

$$\beta^2 k > 0, \quad t > 0$$

A_n are constants.

As before then we get that for every $n \geq 1$

$$u_n(x, t) = \tilde{A}_n e^{-\left(\frac{n\pi}{L}\right)^2 kt} \cdot \sin\left(\frac{n\pi}{L} x\right) \quad (10)$$

is a solution of

$$\begin{cases} u_t - k u_{xx} = 0 & 0 < x < L; t > 0 \\ u(0, t) = 0 = u(L, t) & \forall t > 0 \end{cases}$$

As before

$$u_N(x, t) = \sum_{n=1}^N u_n(x, t) \quad \text{is}$$

also a solution.

Then we still need to match the given initial condition $u(x, 0) = \phi(x)$. So again the question is when and how we have

$$u_N(x, 0) = \sum_{n=1}^N u_n(x, 0)$$
$$= \sum_{n=1}^N \tilde{A}_n \sin\left(\frac{n\pi}{L}x\right) \stackrel{?}{=} \phi(x) \quad (*)$$

Same as with the case of the wave we'll see using Fourier analysis/series that any ϕ continuous L -periodic can be

be written as a (possible infinite)

sine sum (sine series) as in $\textcircled{\neq}$

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To finish the zero Dirichlet case for wave and heat we need to prove our claim that only $\lambda > 0$ is possible.

To prove this we will show that if $\lambda = 0$, $\lambda < 0$ or λ is complex we have no solutions other than the trivial solution to the boundary value problem

with zero Dirichlet boundary conditions for the wave or the heat/diffusion equations (13)

① Let's suppose $\lambda = 0$; then the second order equation for X is $X''(x) = 0$ (for either wave or heat) which implies $X(x) = Ax + B$ for some constants A, B . But the zero Dirichlet boundary conditions imply $0 = X(0) = B$ so $B = 0$; while $0 = X(L) = A \cdot L$; but $L \neq 0 \Rightarrow A = 0$. So $X(x) \equiv 0$ for all

$$0 < x < L. \Rightarrow u(x,t) = X(x)T(t) \equiv 0.$$

(14)

for all $x \in [0, L]$ and $t > 0$.

② Let's suppose $\lambda < 0$: Then we can write

$$\lambda = -(\gamma^2) \text{ with } \gamma > 0 \text{ (say } \gamma = \sqrt{-\lambda} \text{)}.$$

Then, $X''(x) = \gamma^2 X(x)$ has solution

$$X(x) = C \cosh(\gamma x) + D \sinh(\gamma x).$$

Recall $\cosh(\gamma x) = \frac{e^{\gamma x} + e^{-\gamma x}}{2}$; $\sinh(\gamma x) = \frac{e^{\gamma x} - e^{-\gamma x}}{2}$.

Now, let's use the zero Dirichlet boundary conditions:

(5)

$$0 = X(0) = C \cosh(0) \Rightarrow C = 0.$$

$$0 = X(L) = D \underbrace{\sinh(\gamma L)}_{\substack{= 1 \\ \neq 0}} \Rightarrow D = 0$$

So $X(x) = 0$, for all $x \in [0, L]$

$\sinh(y) = 0$ if and only if $y = 0$.
But $\gamma \neq 0, L \neq 0$.

$\Rightarrow$ (as before) $u(x, t) \equiv 0$ for all $x \in [0, L], t > 0$.

Again we only get the trivial solution!

③ Suppose λ is a complex number $\lambda = a + ib$

Let $\gamma = \alpha + i\delta$ be a complex number such

that $\lambda = (\gamma^2) = (-\gamma)^2$. Such γ always

exists. To see this write λ in polar coordinates (16)

$$\lambda = \rho e^{i\theta}; \quad \rho > 0 \quad 0 < \theta < 2\pi. \text{ then}$$

$$\gamma = \sqrt{\lambda} \text{ is defined by } \gamma = \sqrt{\rho} e^{i\theta/2}$$

Then our second order ODE for X reads as

$$X''(x) = -\lambda X(x) (= \gamma^2 X(x) = (-\gamma^2) X(x))$$

$$\text{So } X(x) = C e^{\gamma x} + D e^{-\gamma x} \quad (\text{recall } \gamma \text{ is complex!})$$

Now our zero Dirichlet boundary conditions say:

$$0 = X(0) = C + D \Rightarrow \boxed{C = -D}$$

$$0 = X(L) = -D e^{\gamma L} + D e^{-\gamma L} \Rightarrow$$

$$D e^{-\gamma L} (-2 e^{2\gamma L} + 1) = 0 \quad (17)$$

Hence either $D e^{-\gamma L} = 0$ which $\Rightarrow D = 0$

or $-2 e^{2\gamma L} + 1 = 0 \Rightarrow e^{2\gamma L} = 1 \quad (+)$

Let's write $\gamma = \alpha + i\delta$; then from $(+)$ we

get that $e^{2\gamma L} = \underbrace{e^{2\alpha L} e^{2i\delta L}}_{(+)} = 1$. But this can

only happen if $e^{2\alpha L} = 1$ (recall α is real)

which implies $\alpha = 0$ since $L \neq 0$ and $\delta = \frac{\pi n}{L}$

(since $e^{2i\delta L} = \cos(2\delta L) + i \sin(2\delta L)$). $\Rightarrow$ all in

all we get that $\gamma = \frac{i\pi n}{L}$ for $n \geq 1$. (18)

But $-1 = \gamma^2 = \left(\frac{i\pi n}{L}\right)^2 = -\left(\frac{\pi n}{L}\right)^2$ which

is a Real number while 1 (and hence -1)

was assumed to be complex! We get then

a contradiction. Thus 1 cannot be complex.

From (1), (2), (3) we get that $1 > 0$ is the

only possibility for nontrivial solutions

as we had claimed.

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